CN LAB
ASSIGNMENT10

Nithin S

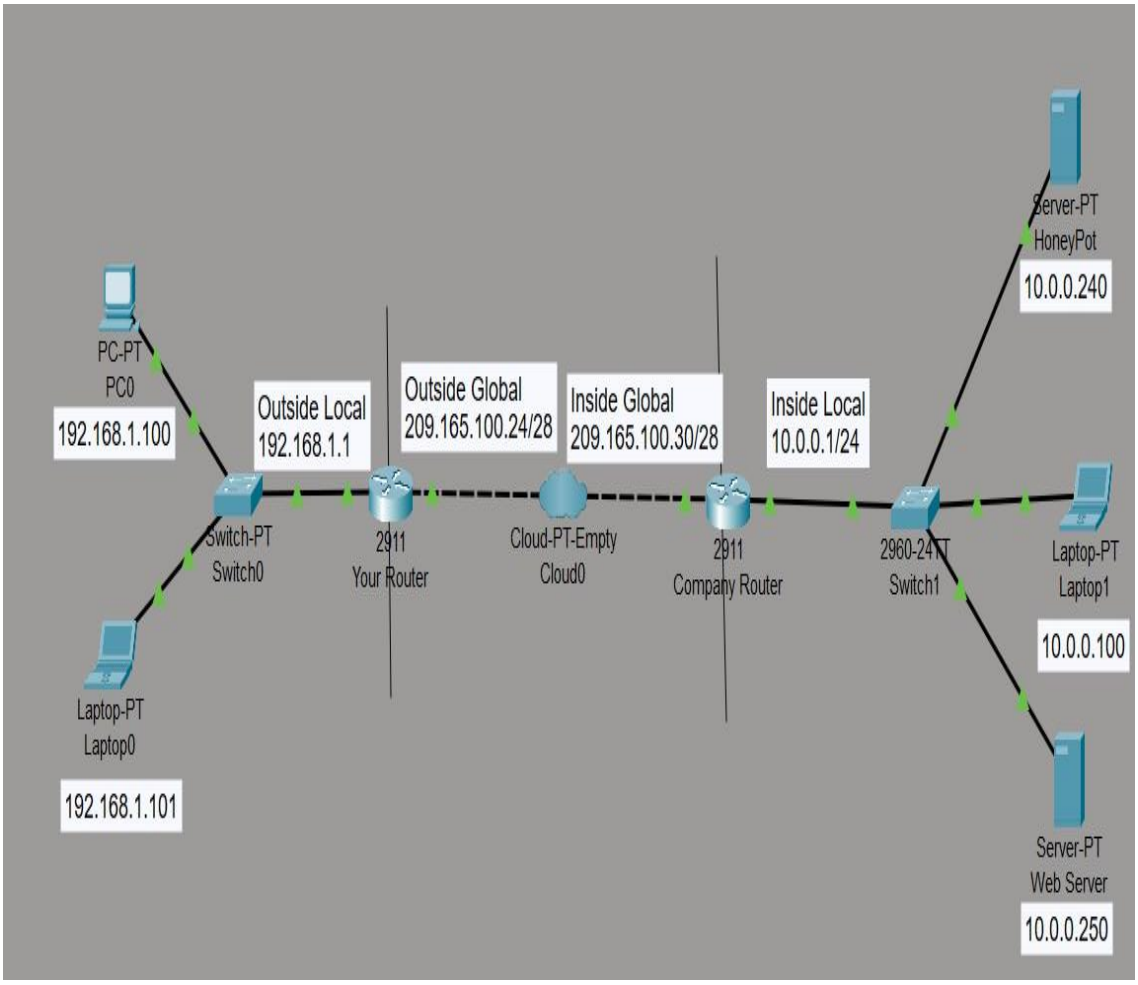
24BCE1319

L57+L58

Network Address Translation (NAT) Implementation

Consider the following topology:

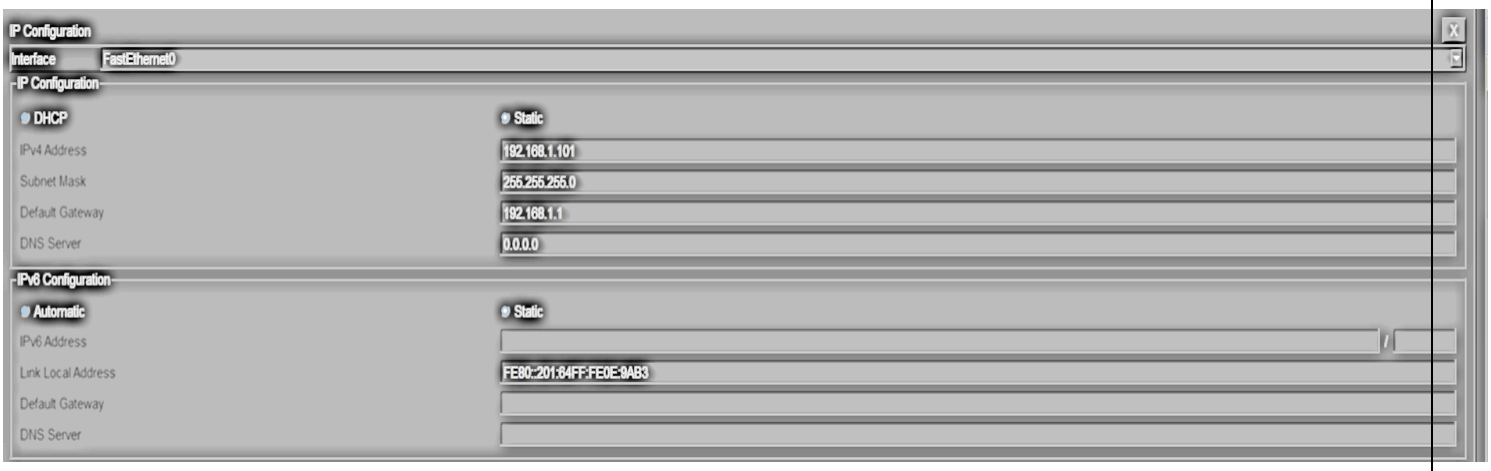
With reference to the attached documents, perform Network Address Translation for a different topology by configuring and testing Static NAT, Port Forwarding, PAT, and Dynamic NAT. Include the required router/device configurations and the corresponding output verification for each case.



Configurations:

PC0:

Laptop0:



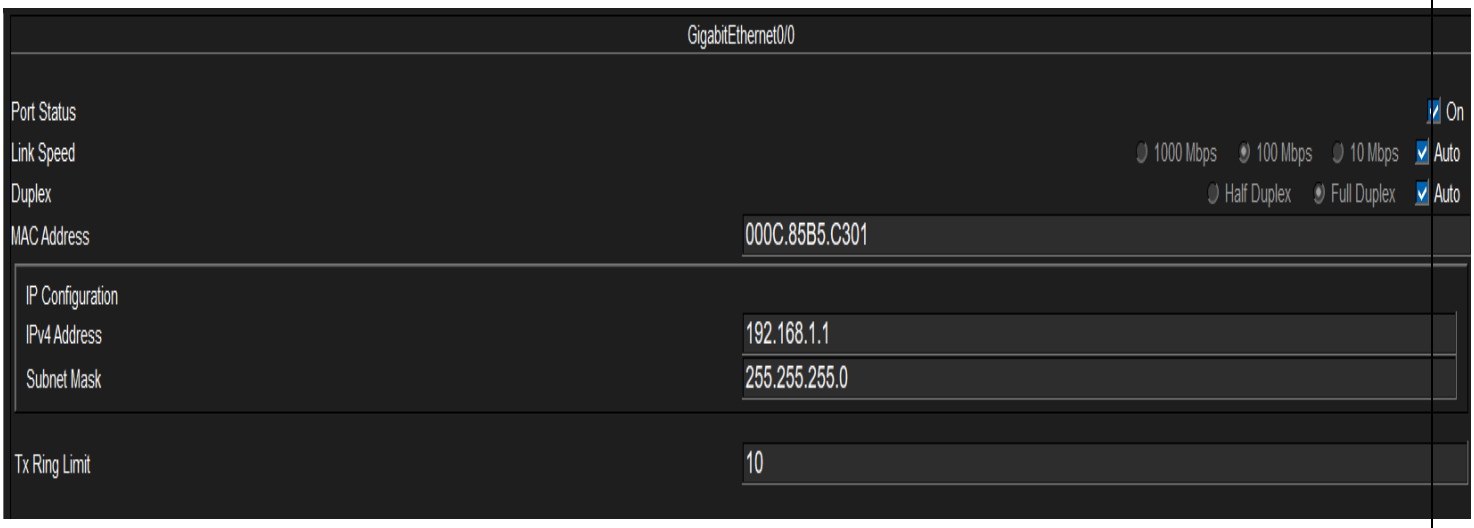
HoneyPot:

Laptop1:

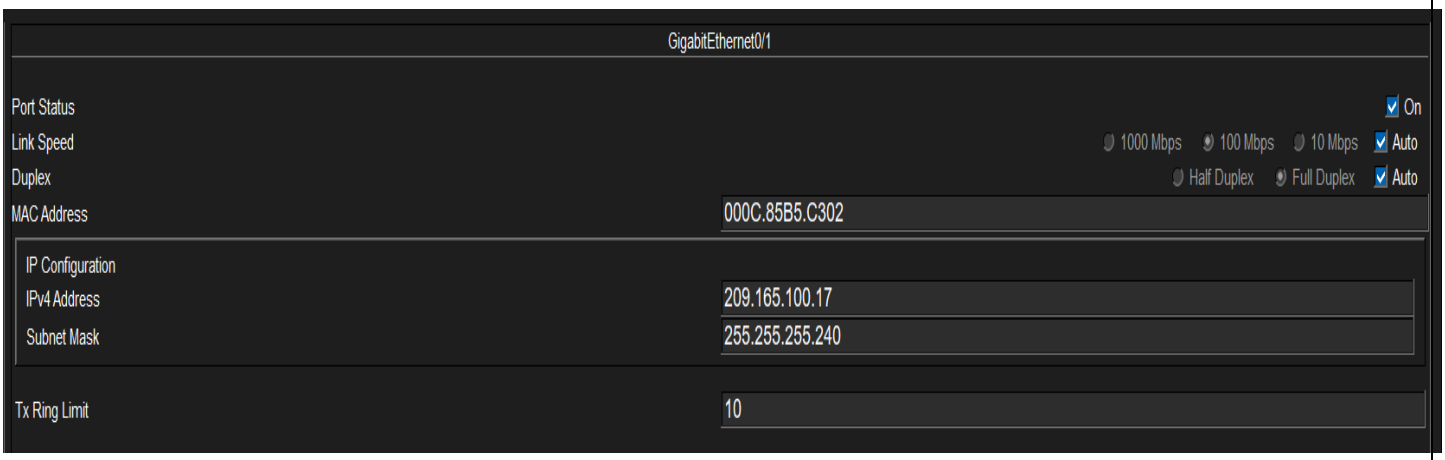
Web Server:

Your Router:

GigabitEthernet0/0:

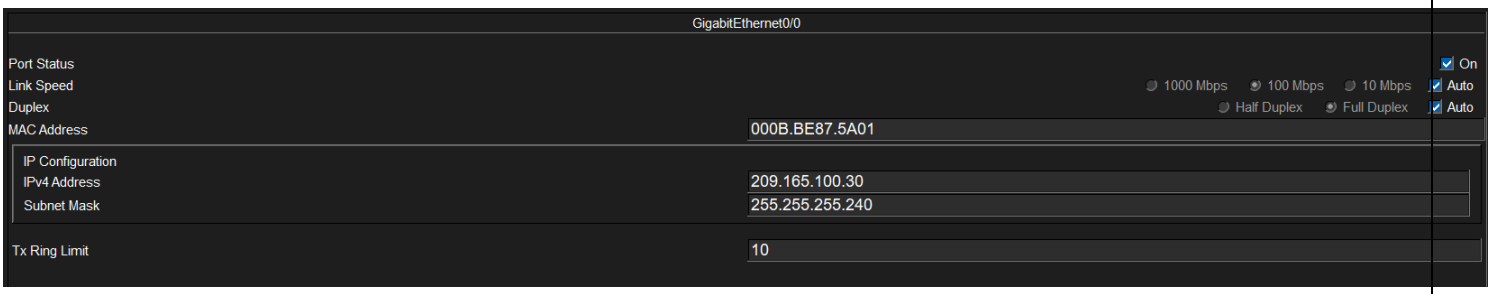


GigabitEthernet0/1:

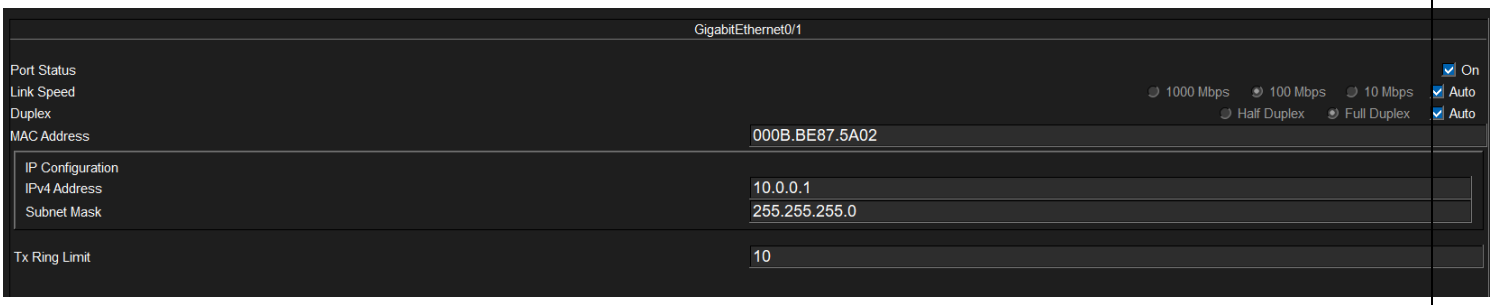


Company Router:

GigabitEthernet0/0:



GigabitEthernet0/1:



(a) Static NAT:

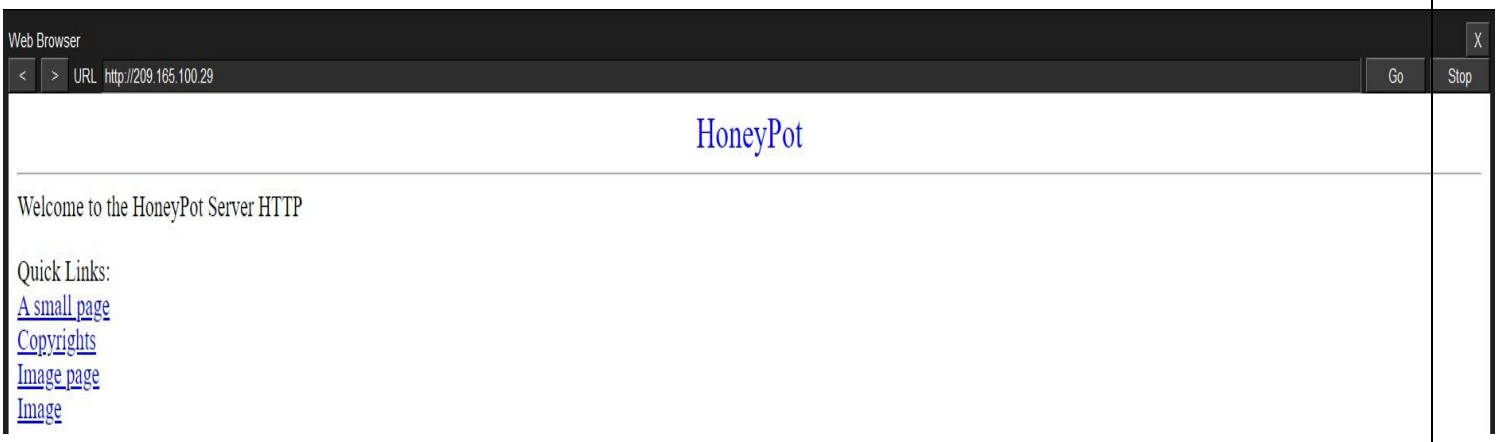
Config for Company Router:

```
company>en
company#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
company(config)#int g0/0
company(config-if)#ip nat outside
company(config-if)#exit
company(config)#int g0/1
company(config-if)#ip nat inside
company(config-if)#exit
company(config)#ip nat inside source static 10.0.0.240 209.165.100.29
company(config)#ip route 192.168.1.0 255.255.255.0 209.165.100.17
company(config)#end
company#
```

Config for Your Router:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname yourRouter
yourRouter(config)#ip route 10.0.0.0 255.255.255.0 209.165.100.30
yourRouter(config)#end
yourRouter#
```

Output from Laptop0 when accessing HoneyPot's HTTP Services:

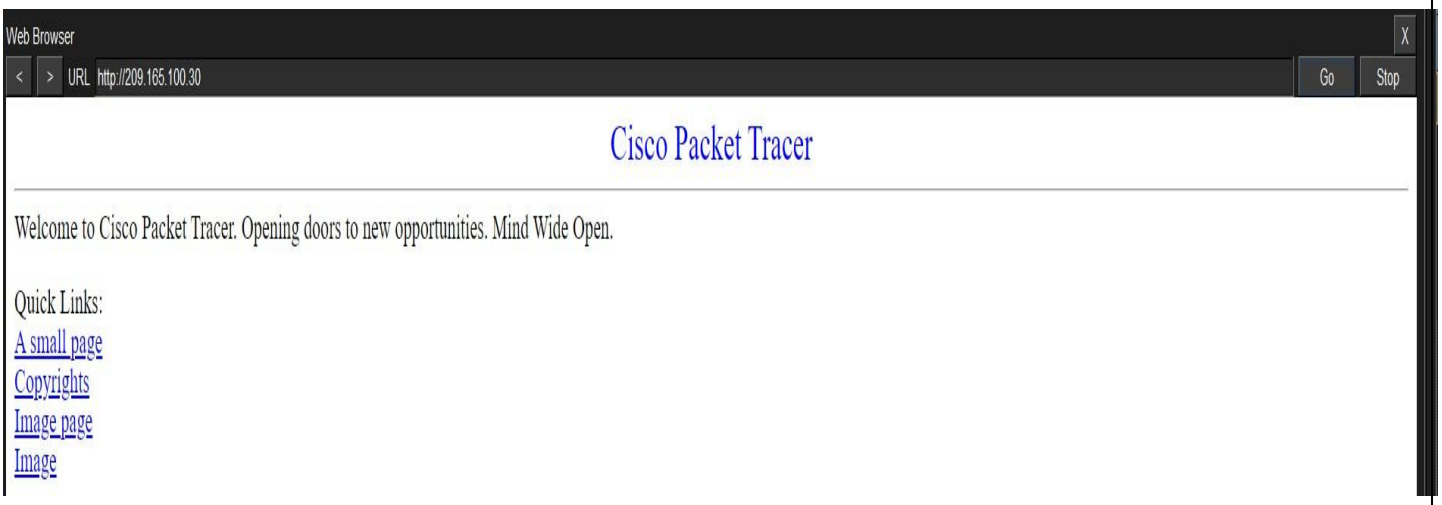


(b) Port forwarding:

Company router config:

```
company#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
company(config)#ip nat inside source static tcp 10.0.0.250 80 209.165.100.30 80
company(config)#end
company#
```

Output from PC0 when Accessing Web Server's HTTP Services via Port 80:



(c) PAT (Port Address Translation):

Company Router Configuration:

```
company#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
company(config)#access-list 10 permit 10.0.0.0 0.0.0.255
company(config)#ip nat inside source list 10 interface g0/0 overload
company(config)#end
company#
```

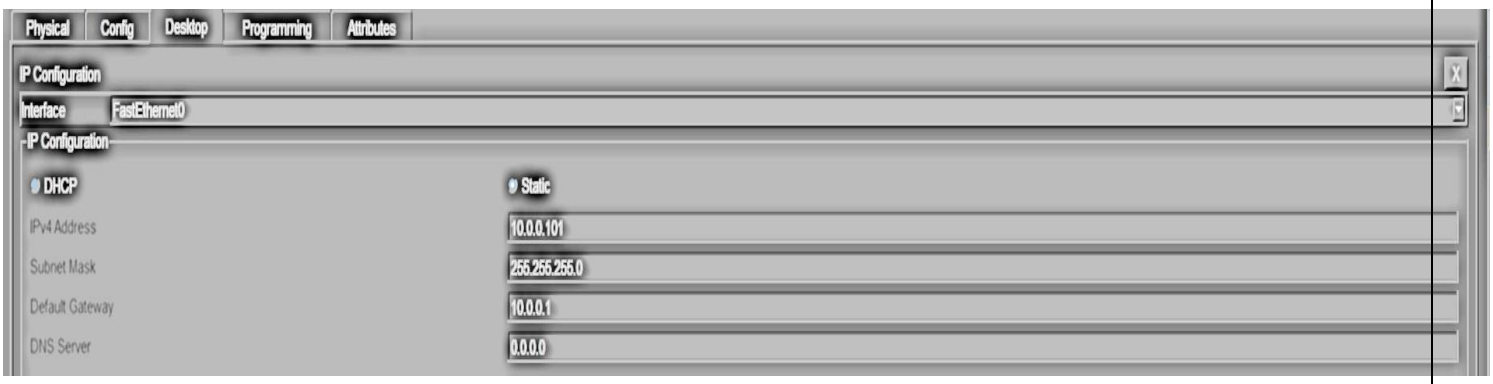
Ping result from Laptop1:

```
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time=7ms TTL=128
Reply from 192.168.1.101: bytes=32 time=7ms TTL=128
Reply from 192.168.1.101: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 5ms
```

PAT result from Company Server:

```
company#show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 209.165.100.29:21 10.0.0.240:21      192.168.1.101:21   192.168.1.101:21
icmp 209.165.100.29:22 10.0.0.240:22      192.168.1.101:22   192.168.1.101:22
---  209.165.100.29    10.0.0.240         ---                ---
tcp 209.165.100.29:80  10.0.0.240:80      192.168.1.101:1025 192.168.1.101:1025
tcp 209.165.100.29:80  10.0.0.240:80      192.168.1.101:1026 192.168.1.101:1026
tcp 209.165.100.30:80  10.0.0.250:80      ---                ---
tcp 209.165.100.30:80  10.0.0.250:80      192.168.1.100:1025 192.168.1.100:1025
```

After Laptop1 IP Reconfiguration:



Ping result from Laptop1:

```
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126
Reply from 192.168.1.100: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PAT Result from Company Server:

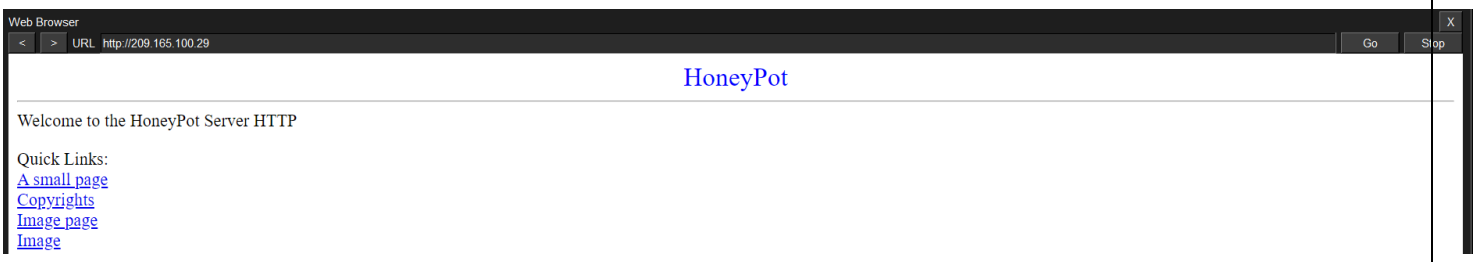
```
company#show ip nat translation
Pro  Inside global     Inside local      Outside local     Outside global
icmp 209.165.100.30:10 10.0.0.166:10     192.168.1.100:10  192.168.1.100:10
---  209.165.100.29    10.0.0.240        ---               ---
tcp 209.165.100.29:80  10.0.0.240:80     192.168.1.101:1025 192.168.1.101:1025
tcp 209.165.100.29:80  10.0.0.240:80     192.168.1.101:1026 192.168.1.101:1026
tcp 209.165.100.30:80  10.0.0.250:80     ---               ---
tcp 209.165.100.30:80  10.0.0.250:80     192.168.1.100:1025 192.168.1.100:1025
```

(d) Dynamic NAT:

YourRouter Config:

```
yourRouter>en
yourRouter#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
yourRouter(config)#int g0/0
yourRouter(config-if)#ip nat inside
yourRouter(config-if)#end
yourRouter#
%SYS-5-CONFIG_I: Configured from console by console

yourRouter#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
yourRouter(config)#int g0/1
yourRouter(config-if)#ip nat outside
yourRouter(config-if)#exit
yourRouter(config)#access-list 1 permit 192.168.1.0 0.0.0.255
yourRouter(config)#ip nat pool MYPOOL 209.165.100.17 209.165.100.24 netmask 255.255.255.240
yourRouter(config)#ip nat inside source list 1 pool MYPOOL overload
yourRouter(config)#
```

HoneyPot server access from PC0:



NAT Translations from YourRouter:

```
yourRouter#show ip nat translations
Pro  Inside global      Inside local        Outside local       Outside global
tcp 209.165.100.17:1026192.168.1.100:1026 209.165.100.29:80  209.165.100.29:80
```